

CUSTOMER 360 INTELLIGENCE

Brazilian E-Commerce Customer Analytics & Retention Dashboard

Power BI Data Analytics Portfolio Project

Dataset Source: Kaggle

Original Dataset: Brazilian E-Commerce Public Dataset by Olist

Tools: Power BI • Power Query • DAX

Year: 2026

1. EXECUTIVE SUMMARY

Customer 360 Intelligence is a Power BI analytics project developed using the Brazilian E-Commerce Public Dataset by Olist.

The dataset was downloaded from Kaggle and transformed into an analytical data model using Power Query, Power BI and DAX.

The primary objective of the project is to provide a 360-degree view of e-commerce business performance and customer Behaviour.

The dashboard combines sales, customer, product, seller, payment and review data to Analyse:

Revenue performance

Order performance

Product and seller contribution

Customer purchasing Behaviour

One-time and repeat customers

Customer churn

Customer satisfaction

Customer value

Revenue at risk

Retention and growth opportunities

Rather than simply presenting historical sales figures, the project focuses on converting data into business insights and actionable recommendations.

2. BUSINESS PROBLEM

An e-commerce business needs to understand more than its total sales.

Management needs visibility into:

What products and sellers are driving sales?

Which states contribute the most revenue?

How many customers return after their first purchase?

What percentage of customers are one-time buyers?

Which repeat customers are becoming inactive?

Where is revenue potentially exposed because of customer churn?

Which customer segments have higher value?

How satisfied are customers with products and sellers?

Where should retention and growth efforts be focused?

Without a connected customer-centric analytical model, these questions can be difficult to answer efficiently.

Therefore, this project was developed to combine multiple business areas into a single interactive Customer 360 Intelligence dashboard.

3. PROJECT OBJECTIVES

3.1 Sales Performance

Understand overall revenue, order volume, product performance, seller performance and state-level sales.

3.2 Customer Behaviour

Analyse customer purchasing patterns and compare one-time and repeat customers.

3.3 Customer Retention

Identify repeat customers, churned customers and opportunities to increase repeat purchasing.

3.4 Customer Satisfaction

Analyse customer reviews, satisfaction levels and positive/negative review rates across categories and sellers.

3.5 Customer Value

Identify higher-value customers and understand how customer value changes with purchase frequency.

3.6 Growth Opportunities

Identify areas where the business can improve retention, increase purchase frequency and protect revenue.

4. DATASET

Brazilian E-Commerce Public Dataset by Olist

The dataset used in this project was downloaded from Kaggle.

The dataset is based on the Brazilian E-Commerce Public Dataset by Olist, which contains information about orders placed on the Olist marketplace in Brazil.

The dataset provides information across multiple business areas, including:

Customers

Orders

Order Items

Payments

Products

Sellers

Reviews

Geographical information

Approximate Dataset Scale

Metric, Approximate Volume

Orders, ~100K

Unique Customers, ~99K

Products Sold, ~113K

Reviews, ~98K

The dataset provides a strong foundation for analyzing e-commerce sales, customers, products, sellers, payments and customer satisfaction.

5. DATA SOURCE

Dataset Source

Kaggle

Original Dataset

Brazilian E-Commerce Public Dataset by Olist

The Kaggle dataset was downloaded and imported into Power BI for transformation and analysis.

The original dataset was not modified directly. Instead, Power Query transformations and additional Power BI structures were used to prepare the data for analytical reporting.

6. DATA PREPARATION

The raw dataset was prepared using Power Query before being used for dashboard development.

The preparation process included:

Data type validation

Data cleaning

Column selection

Data transformation

Relationship preparation

Customer-level data preparation

Product and category preparation

Date preparation

Structuring tables for analytical use

Data Quality Issues Identified

During the data preparation process, two data quality issues were identified and documented in the Data Quality Log:

DQ-001 — Missing Delivered Dates

In Fact_Order, 8 rows had no order_delivered_customer_date even though the order status was recorded as Delivered. The issue was classified as Low Severity because it may affect delivery lead-time analysis. The issue was documented, while the source data was retained.

DQ-002 — Missing Product Categories

In Dim_Products, the product_category_name field was missing for approximately 3% of products (623 products). This affected 1,627 order-item records across 1,473 unique orders. The issue was classified as Medium Severity because category-level sales and order analysis could show blank categories. The affected products were classified as “Uncategorized” for analysis while retaining the original source data.

The identified data quality issues and their corresponding actions were recorded in the Data Quality Log. The prepared data was then loaded into Power BI and organized into a connected analytical model for sales, customer, product, payment, seller, satisfaction, and retention analysis.

7. DATA MODEL

The project uses a multi-table analytical data model connecting customers, orders, products, sellers, payments and reviews.

Original Dataset Tables

The main source tables from the Kaggle dataset include:

Customers

Orders

Order Items

Order Payments

Order Reviews

Products

Sellers

Geolocation

These tables represent the original business data provided by the dataset.

8. TABLES CREATED IN POWER BI

Two important supporting tables were created specifically during this project.

These tables were not part of the original Kaggle dataset.

8.1 Dim_UniqueCustomer

Dim_UniqueCustomer was created to provide a unique customer-level analytical dimension based on customer_unique_id.

The original customer data can contain multiple customer IDs associated with the same underlying customer.

The unique customer dimension was therefore created to support customer-centric analysis.

It is used for:

Unique customer analysis

One-time customer classification

Repeat customer classification

Purchase frequency

Customer churn

Customer value

Retention analysis

This table was created as part of the Power BI project and was not an original table in the Kaggle dataset.

8.2 Measure

A separate Measure table was created specifically to organize the DAX measures developed for the dashboard.

This table helps keep the analytical calculations organized and separate from the source data tables.

The measures cover areas such as:

Revenue

Orders

Customers

Customer value

Repeat customers

Churn

Revenue at risk

Reviews

Product performance

Seller performance

Retention opportunities

The Measure table was created in Power BI and was not part of the original Kaggle dataset.

9. DATA MODEL STRUCTURE

The main analytical flow can be summarized as:

Customers → Orders → Order Items → Products / Categories

Additional connections support:

Orders → Payments

Orders → Reviews

Order Items → Sellers

This structure allows the dashboard to Analyse the same business activity from multiple perspectives.

10. CALCULATED COLUMNS CREATED

Additional calculated columns were created in Power BI to support customer segmentation and analytical classification.

These columns were not provided directly by the Kaggle dataset.

10.1 Purchase Type

Customers are classified into:

No Orders

One-Time

Repeat

This classification allows the dashboard to compare customer Behaviour between first-time and returning buyers.

10.2 Customer State

A customer state field was created to associate the unique customer dimension with the corresponding customer geographical information.

This allows customer analysis to be performed at the state level.

10.3 Churn Status

A churn classification was created to identify repeat customers who have not made a purchase for more than 180 days based on their latest purchase date in the dataset.

This classification is used to identify customers who are no longer actively purchasing and may require retention efforts.

10.4 Order Frequency Group

Customers are grouped according to purchase frequency.

The groups include:

1 Order

2 Orders

3 Orders

4–5 Orders

6–10 Orders

10+ Orders

This allows the project to understand how customer value changes as purchase frequency increases.

10.5 Frequency Sort

A supporting sort of column was created to ensure the order-frequency groups appear in logical business order rather than alphabetical order.

10.6 Satisfaction Level

Review scores were transformed into meaningful satisfaction categories based on the **1–5 review score range**:

Review Score	Satisfaction Level
1	Very Dissatisfied
2	Dissatisfied
3	Neutral
4	Satisfied
5	Very Satisfied

Review Score Range: 1 = Very Dissatisfied → 5 = Very Satisfied

This classification makes the numerical review scores easier to interpret for business and customer satisfaction analysis.

11. DAX ANALYTICS

DAX was used extensively to create reusable business metrics and analytical calculations.

The project includes custom measures for:

Total Revenue

Total Orders

Total Customers

Active Customers

Revenue Growth

Customer Revenue

Average Customer Value

Repeat Customers

Active Repeat Customers

One-Time Customers

Churned Customers

Churn Rate

Revenue at Risk

High-Value Customers

Repeat Customer Revenue

Category Revenue

Category Orders

Category Revenue per Order

Total Reviews

Average Review Score

Positive Reviews

Negative Reviews

Positive Review Rate

Negative Review Rate

Seller Review Performance

These measures were created specifically for this project and were not supplied by the Kaggle dataset.

12. IMPORTANT KPI DEFINITIONS

12.1 Total Revenue

Total revenue is calculated using payment value from the order payment table.

Total Revenue =

SUM(Fact_OrderPayments[payment_value])

The dashboard shows approximately 16.0M in payment-based revenue.

12.2 Product Revenue

Product revenue is calculated from item-level product prices.

Category Revenue =
 $\text{SUM}(\text{Fact_OrderItems}[\text{price}])$

The project analysis shows approximately 13.56M in product revenue.

The difference between payment-based revenue and product revenue is expected because the payment table and item-price table represent different aspects of the transaction.

13. CUSTOMER ANALYTICS DEFINITIONS

13.1 Repeat Customers

Repeat Customers are calculated as:

Active Repeat Customers + Churned Customers

Repeat Customers =
 $[\text{Active Repeat Customers}] + [\text{Churned Customers}]$

This ensures that customers who previously made multiple purchases but are now classified as churned are still recognized as repeat customers.

14. CHURN DEFINITION

For this project, churn is defined using a 180-day inactivity rule.

A repeat customer is classified as churned when there has been no purchase for more than 180 days from the latest purchase date represented in the dataset.

This is a rule-based business definition.

It is not a machine-learning churn prediction model.

The purpose is to provide a practical retention indicator using the historical data available.

15. REVENUE AT RISK

Revenue at Risk represents the historical revenue generated by customers classified as churned.

Revenue at Risk =

VAR ChurnedCustomers =

```
CALCULATETABLE(  
    VALUES(Dim_UniqueCustomer[customer_unique_id]),  
    Dim_UniqueCustomer[Churn Status] = "Churned"  
)
```

RETURN

```
CALCULATE(  
    [Customer Revenue],  
    TREATAS(  
        ChurnedCustomers,  
        Dim_Customers[customer_unique_id]  
    )  
)
```

Important Interpretation

Revenue at Risk should be interpreted as:

Historical revenue exposure associated with churned customers.

It should not be interpreted as a guaranteed future revenue loss or a predictive forecast.

16. HIGH-VALUE CUSTOMERS

For this project, customers generating ₹2,000 or more in customer revenue are classified as high-value customers.

High-Value Customers =

```
COUNTROWS(  
    FILTER(  
        VALUES(Dim_UniqueCustomer[customer_unique_id]),  
        [Customer Revenue] >= 2000  
    )  
)
```

This segmentation helps identify customers who may deserve additional retention and loyalty attention.

17. REPEAT CUSTOMER REVENUE

Repeat Customer Revenue measures the revenue generated by customers classified as repeat customers.

Repeat Customer Revenue =

```
CALCULATE(  
    [Customer Revenue],  
    Dim_UniqueCustomer[Purchase Type] = "Repeat"  
)
```

This metric helps quantify the financial contribution of repeat customers.

18. AVERAGE CUSTOMER VALUE

Average Customer Value is calculated as customer revenue divided by the number of unique customers in the applicable context.

Average Customer Value =

```
DIVIDE(  
    [Customer Revenue],  
    DISTINCTCOUNT(Dim_UniqueCustomer[customer_unique_id]),  
    0  
)
```

This measure is used to understand how customer value changes across purchase-frequency groups.

19. DASHBOARD STRUCTURE

The final Power BI solution contains five analytical pages.

PAGE 1 — EXECUTIVE MISSION CONTROL

Purpose

Provides an executive overview of the business.

Key Metrics

Total Revenue

Total Orders

Total Customers

Active Customers

Revenue Growth

Key Visuals

Monthly Sales & Revenue Trend

Revenue by State

Payment Method Distribution

Review Status

Customer Activity

Business Value

This page provides management with a quick view of overall business performance and allows users to identify changes in revenue, orders, customer activity and regional performance.

PAGE 2 — SALES ANALYTICS

Purpose

Provides detailed analysis of product, category and seller performance.

Key Metrics

Product Categories

Total Sellers

Product Revenue

Top Sold Product Category

Key Visuals

Top 10 Best-Selling Products

Top 10 Sellers by Sales

Sales Performance Table

Category Review Sentiment

Business Value

This page helps identify major sales contributors and provides insight into product, category and seller performance.

PAGE 3 — CUSTOMER ANALYTICS

Purpose

Analyses customer Behaviour and retention.

Key Metrics

Repeat Customers

Active Repeat Customers

One-Time Customers

Churned Customers

Churn Rate

Key Visuals

Customer Churn Overview

Top 10 States by Churned Customers

Customer Purchase Behaviour

Customer Revenue by Churn Status

Customer Purchase Frequency

Business Value

This page helps the business understand customer lifecycle Behaviour and identify areas where retention strategies are required.

PAGE 4 — CUSTOMER SATISFACTION DISTRIBUTION

Purpose

Analyses customer satisfaction using review data.

Key Metrics

Total Reviews

Average Review Score

Positive Reviews

Negative Reviews

Key Visuals

Top 10 Categories by Positive Review Rate

Top 10 Categories by Negative Review Rate

Seller Review Performance

Customer Satisfaction Distribution

Business Value

This page highlights satisfaction patterns across categories and sellers and helps identify potential product or service improvement areas.

PAGE 5 — CUSTOMER VALUE & RETENTION STRATEGY

Purpose

Transforms customer Behaviour analysis into actionable retention and growth opportunities.

Key Metrics

Revenue at Risk

High-Value Customers

Repeat Customer Revenue

Repeat Purchase Opportunity

Key Visuals

Customer Value & Loyalty Matrix

Revenue at Risk by State

Average Customer Value by Order Frequency

Revenue Concentration by Customer Frequency

High-Value Customer Analysis

Business Value

This page helps management prioritize retention efforts based on customer value, purchase Behaviour and revenue exposure.

20. KEY BUSINESS INSIGHTS

INSIGHT 1 — ONE-TIME CUSTOMERS REPRESENT A MAJOR OPPORTUNITY

The Olist dataset contains a very high proportion of one-time buyers.

This indicates that a major growth opportunity lies in converting existing customers into repeat buyers.

Instead of focusing only on acquiring new customers, the business can potentially increase revenue by encouraging existing customers to make a second purchase.

INSIGHT 2 — PURCHASE FREQUENCY IS LINKED TO CUSTOMER VALUE

Customers with multiple purchases generally contribute more cumulative revenue than one-time buyers.

This indicates that increasing purchase frequency can be an important long-term growth strategy.

The customer-frequency analysis helps identify opportunities to move customers from:

First Purchase → Second Purchase → Repeat Customer → Higher-Value Customer

INSIGHT 3 — CHURN CREATES REVENUE EXPOSURE

Revenue-at-risk analysis highlights historical revenue generated by customers classified as churned.

This provides a way to identify where retention efforts could have the greatest financial impact.

States with higher revenue exposure can be prioritized for targeted retention strategies.

INSIGHT 4 — CUSTOMER SATISFACTION IS GENERALLY STRONG

The dashboard shows an average review score of approximately:

4.10 / 5

Approximately:

76K Positive Reviews

14K Negative Reviews

98K Total Reviews

Although overall satisfaction is strong, category-level and seller-level differences provide opportunities for targeted improvement.

INSIGHT 5 — SALES ARE CONCENTRATED AMONG LEADING PRODUCTS AND SELLERS

The Top 10 product and seller analysis identifies major contributors to sales performance.

Understanding these high-performing contributors can help management focus on:

High-performing products

Important seller relationships

High-value categories

Commercial opportunities

21. BUSINESS RECOMMENDATIONS

21.1 Convert One-Time Buyers

Develop targeted post-purchase campaigns to encourage a second purchase.

Potential strategies include:

Personalized offers

Product recommendations

Second-purchase incentives

Follow-up communication

The objective should be to move customers from one-time purchase to repeat purchase.

21.2 Protect High-Value Customers

Customers generating higher revenue should receive greater retention attention.

Potential strategies include:

Loyalty programs

Personalized promotions

Exclusive offers

Early access to products

Customer-specific recommendations

21.3 Prioritize High-Risk States

Use the Revenue at Risk by State analysis to identify geographical areas with higher historical revenue exposure from churned customers.

Retention campaigns can then be prioritized based on potential financial impact.

21.4 Increase Purchase Frequency

Customer frequency segments can be used to create targeted campaigns.

For example:

1 Order → Encourage 2nd Order

2–3 Orders → Encourage Continued Purchasing

4+ Orders → Strengthen Loyalty

This creates a more targeted customer retention strategy.

21.5 Monitor Negative Review Rates

Categories and sellers with higher negative-review rates should be investigated.

Possible causes may include:

Product quality

Seller service

Delivery experience

Product expectations

Customer support

Addressing recurring issues can improve satisfaction and potentially support repeat purchasing.

22. TECHNOLOGY USED

Power BI

Power BI was used for:

Data modeling

Dashboard development

Interactive visualizations

KPI creation

Cross-filtering

Business intelligence reporting

Power Query

Power Query was used for:

Data cleaning

Data transformation

Data preparation

Structuring source tables

Preparing data for analysis

DAX

DAX was used for:

Business KPIs

Customer segmentation

Customer value analysis

Churn calculations

Revenue-at-risk analysis

Review analysis

Product and seller metrics

Retention analysis

23. PROJECT DEVELOPMENT PROCESS

The project followed the following analytical workflow:

STEP 1 — Dataset Acquisition

The Brazilian E-Commerce Public Dataset by Olist was downloaded from Kaggle.

↓

STEP 2 — Data Preparation

Source tables were cleaned and transformed using Power Query.

↓

STEP 3 — Data Modeling

Relationships were established between customers, orders, order items, products, sellers, payments and reviews.

↓

STEP 4 — Analytical Tables

Additional tables were created:

Dim_UniqueCustomer

Measure

↓

STEP 5 — Customer Segmentation

Calculated columns were created for:

Purchase Type

Customer State

Churn Status

Order Frequency Group

Frequency Sort

Satisfaction Level

↓

STEP 6 — DAX Development

Custom measures were created for revenue, customers, orders, churn, reviews, customer value and retention.

↓

STEP 7 — Dashboard Development

Five Power BI pages were developed:

Executive Mission Control

Sales Analytics

Customer Analytics

Customer Satisfaction Distribution

Customer Value & Retention Strategy

↓

STEP 8 — Business Interpretation

The analytical results were translated into insights and business recommendations.

24. PROJECT LIMITATIONS

Historical Data

The analysis is based on historical Olist data and does not represent a live production customer database.

Rule-Based Churn

Churn is calculated using a 180-day inactivity rule rather than a machine-learning prediction model.

Revenue at Risk

Revenue at Risk represents historical revenue generated by churned customers.

It should not be interpreted as an exact forecast of future revenue loss.

Dataset Characteristics

The very high proportion of one-time customers is a characteristic of the Olist dataset.

This should be considered when interpreting customer retention metrics.

Data Relationships

Some analyses require cross-table filtering through order IDs because customers, orders, payments, products, sellers and reviews are stored in separate tables.

25. PROJECT CONTRIBUTION

The main contribution of this project was not simply creating visualizations from the raw Kaggle dataset.

The project transformed the source data into a customer-centric analytical solution.

Additional analytical structures were created specifically for this project, including:

Dim_UniqueCustomer

Measure

Customer segmentation columns

Churn classification

Purchase-frequency groups

Satisfaction classification

Custom DAX measures

These additions enabled analysis that was not directly available in the original dataset.

The final solution therefore provides deeper insight into:

Customer lifecycle

Purchase Behaviour

Customer value

Churn

Revenue exposure

Satisfaction

Retention opportunities

26. FINAL BUSINESS IMPACT

The Customer 360 Intelligence dashboard enables business users to move from:

PERFORMANCE

What happened?

↓

ANALYSIS

Why is it happening?

↓

CUSTOMER UNDERSTANDING

Who are the customers creating value or risk?

↓

ACTION

Where should the business focus its efforts?

This makes the dashboard more than a reporting solution.

It acts as a decision-support tool for customer retention, sales performance and business growth.

28. CONCLUSION

Customer 360 Intelligence demonstrates the complete data analytics workflow from dataset acquisition to business recommendation.

The project uses the Brazilian E-Commerce Public Dataset by Olist, downloaded from Kaggle, and transforms the raw source data into an interactive Power BI analytical solution.

Through Power Query, Power BI and DAX, the project provides a connected view of:

Sales + Customers + Products + Sellers + Payments + Satisfaction + Retention

The analysis highlights the importance of customer retention, purchase frequency, customer value and satisfaction in driving long-term business growth.

The final dashboard enables decision-makers to identify high-value customers, understand churn, evaluate revenue exposure and prioritize actionable retention opportunities.

CUSTOMER 360 INTELLIGENCE

Turning E-Commerce Data Into Actionable Customer & Business Intelligence

Dataset Source: Kaggle

Original Dataset: Brazilian E-Commerce Public Dataset by Olist

Tools: Power BI • Power Query • DAX

Project Year: 2026